No Country for Indie Developers: A Study of Google Play's Closed Testing Requirements for New Personal Developer Accounts

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In November 2023, Google Play introduced new closed testing requirements for apps submitted by developers operating with personal accounts, or *indie* app developers. These requirements mandate that at least 20 testers must remain opted-in (use the app) for at least 14 consecutive days before the app can be published on the Play Store. According to Google, these new requirements aim to ensure the quality and security of submitted apps. However, for individual developers operating without organizational support, adhering to such requirements can pose logistical challenges and lead to production delays. To understand these challenges, in this paper, we qualitatively analyze app developers' discussions of Google Play's new closed testing requirements on Reddit. Additionally, we report insights from a survey of 14 indie app developers who recently passed the requirements or are actively seeking compliance. Our results show that Google Play's closed testing requirements for indie apps are commonly perceived as discriminatory, imposing logistical and bureaucratic barriers on small-scale creators in their quest to compete in the mobile app market. Our analysis also uncovers the strategies the Android developer community has adapted to navigate such requirements. Based on our findings, we propose several guidelines to help indie app developers integrate the testing requirements into their workflow. We further suggest design strategies to mitigate the impact of such requirements on innovation, fairness, and competition in the mobile app market.

CCS Concepts: • **Human-centered computing → Collaborative and social computing**.

Additional Key Words and Phrases: indie developers, Google Play, closed testing

1 INTRODUCTION

Over the past decade, mobile application (app) stores have grown in size to host millions of apps, offering end-users virtually unlimited options to choose from [1, 56]. To maintain integrity and trustworthiness at such a scale, app stores subject submitted apps to strict reviews, rigorously vetting their security, privacy, and compliance with developer policies. Published apps are also constantly evaluated and monitored for policy violations, such as ranking fraud, user manipulation, or plagiarism [3, 19, 26]. Despite these measures, low-quality apps, such as copycats and malware, frequently appear in app store listings [10, 49, 51].

To defend against the influx of low-quality apps, in November 2023, Google Play enforced a new policy for closed testing, targeting apps submitted by developers operating under personal accounts, or *indie* app developers. The new requirements mandate that indie apps must be tested by 20 users who shall remain opted-in for 14 consecutive days before the *Production Tab* on the Play Store is enabled (Fig. 1). According to Google, these requirements are intended *"to help developers to test their app, identify issues, get feedback, and ensure that everything is ready before they launch"* [17]. However, with no organizational support and a general lack of resources, adhering to Google Play's new testing requirements can substantially disrupt indie app developers' production cycles, delay their time to market, and even prompt a mass migration to corporate structures [18, 45].

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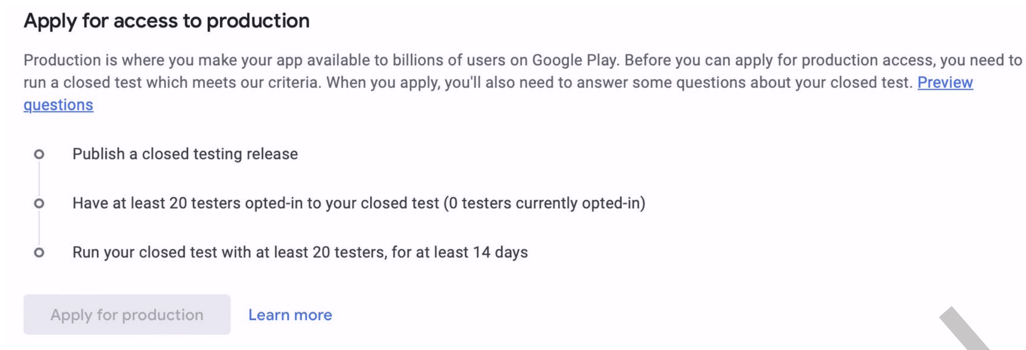

Fig. 1. Google Play's new closed testing requirements for apps submitted by personal developer accounts.

This in turn can diminish autonomy and individual creativity, depriving the ecosystem of a vital stream of entrepreneurial apps [18, 45].

To examine the impact of Google Play's closed testing requirements on current practices, this paper presents a qualitative thematic analysis of developers' Reddit discussions surrounding the policy. Additionally, we survey 14 indie app developers who have either recently met the requirements or are actively working toward compliance. Our findings reveal that indie developers perceive the requirements as frustrating and discriminatory, citing bureaucratic and logistical barriers that hinder their ability to innovate and compete effectively in the mobile app market. We also identify the strategies developers adopt to navigate or circumvent the requirements, including the formation of testing communities and shifting towards alternative software production and distribution models. These insights map the evolving challenges faced by indie developers and highlight the adaptive responses emerging in reaction to restrictive platform policies.

Our findings contribute to the growing body of Software Engineering research that seeks to understand the dynamics of developer communities [6, 7, 28], with a particular focus on indie developers and how platform policies shape their capacity to innovate and compete within one of the world's fastest-growing software ecosystems [13, 22, 36, 43]. By directly engaging with these communities and analyzing their shared experiences, we offer empirical insights that are relevant to both researchers and policymakers. Specifically, this paper makes the following contributions:

- We investigate the short and long-term impacts of Google Play's new closed testing requirements on indie app developer communities. We also explore how these requirements are reshaping the mobile app ecosystem, including the formation of new testing communities, the rise of third-party testing services, and the movement of talent between different mobile app stores and software production paradigms.
- Based on our findings, we propose several research directions to help app stores mitigate the impact of their quality assurance policies on innovation, fairness, and diversity in the mobile app market. We also suggest several guidelines to help indie app developers effectively plan for and adhere to the testing requirements, as well as inform the design and development of similar policies at other competing app stores.

The rest of this paper is organized as follows. In Section 2, we provide background information and formulate our research questions. In Section 3, we present our research method. In Section 4, we discuss our work's implications on the state of research and practice. In Section 5, we address the main limitations of our study. Finally, in Section 6, we conclude the paper.

2 BACKGROUND AND RESEARCH QUESTIONS

As of 2024, low-quality apps continue to pose a major challenge for mobile app distribution platforms [37, 51, 52]. According to the Apple App Store's 2023 Transparency Report [3], out of 6,892,500 apps submitted for review in 2023, 1,763,812 apps were rejected during the screening phase, while 116,117 apps were removed after their initial approval. Similarly, in its 2023 safety report [55], Google Play announced that, in 2022 it blocked 1.43 million low-quality apps and banned 173,000 malicious developer accounts. App stores cite poor design, fraud, safety, and general policy violations as primary reasons for app rejection and removal [26].

To facilitate more effective testing, Apple provides TestFlight, a platform that can be used by developers to distribute pre-release versions of their apps to a select group of beta testers. TestFlight creates a controlled testing environment that helps developers get feedback on the app's functionality, stability, and overall user experience before deployment. Google Play, on the other hand, takes a somewhat different approach to promote pre-release testing, mandating developers with personal developer accounts to recruit 20 users to test their app for 14 consecutive days before the app can be published. This policy effectively pushes part of the quality assurance burden onto developers. The underlying assumption is that a more structured closed testing process can help mitigate abusive app distribution behaviors, such as preventing malicious actors from releasing large numbers of low-quality apps in short periods of time—a persistent issue on Google Play [50]. For instance, in their study of removed Google Play apps, Haoyu et al. [50] reported that several developers with hundreds of published apps had all of their apps removed due to poor quality and reliability. However, enterprise or corporate developer accounts are exempt from these new requirements. This exemption reflects a different trust model within the Google Play ecosystem: enterprise accounts, typically tied to verified business entities, are subject to contractual obligations and enhanced identity verification measures (e.g., DUNS numbers and IRS documentation). These requirements serve as implicit quality signals, offering Google more confidence in the legitimacy and accountability of such developers. However, this differential treatment raises equity concerns. By imposing stricter requirements on personal accounts, Google's policy may unintentionally disadvantage indie developers, burdening them disproportionately compared to their enterprise counterparts.

Over the past decade, indie app developers have significantly shaped the mobile app landscape [36, 42]. Their agility and ability to adapt to the fast-evolving mobile app landscape make them an invaluable asset to the ecosystem. Although often recognized in the gaming sector [13, 30], indie apps are prevalent across all app store categories, particularly in areas considered too risky for larger companies to invest in [36]. According to Freeman and McNeese [15], indie development teams are typically formed around shared aspirations, driven by individual creativity and a collective pursuit of problem-solving and self-improvement. Introducing bureaucratic barriers to these autonomous and passion-driven workflows can lead to the abandonment of such endeavors or a shift towards alternative software production and distribution models. This could result in losing a crucial stream of innovative apps in numerous niche market segments [18, 19, 45].

Existing work on indie software development is often focused on understanding the dynamics of indie game development teams [14, 15, 22, 43], including the challenges they face in the software industry [13, 30, 36] and their ability to adapt new paradigms of software development [31]. However, the research on understanding the complex interplay between platform governance and indie workflows is still underdeveloped. **Our work in this paper bridges this critical gap by uncovering the short and long-term impacts of Google Play's new closed testing requirements on the indie app developer community.** Market research indicates that any industry controlled by a duopoly is likely to be riddled with issues related to market power and barriers to fairness [33]. The app store exemplifies such a duopoly where most apps operate on either Apple's or Google's infrastructures [36]. Therefore, the impact of the new closed testing requirements will likely extend beyond the Android app market to impact competition, creativity, and talent movement in the entire ecosystem. Understanding the effects of platform gatekeeping tactics is essential to help mitigate their impacts as well as inform quality assurance initiatives at other app stores and software distribution platforms. To guide our analysis, we aim to answer the following research questions:

- ***RQ_1: How do indie app developers perceive the new testing requirements?*** Under this research question, we seek to develop an understanding of app developers' perception of Google Play's new closed testing requirements for indie apps, including their potential impacts on fairness, creativity, and competition in the mobile app market.
- ***RQ_2: How do indie app developers navigate the testing requirements?*** Under this research question, we aim to explore the strategies indie app developers employ to bypass or comply with the new testing requirements at individual and community levels. We also seek to identify the pros and cons of these various strategies.

3 METHOD

To answer our research questions, we employ a two-fold qualitative analysis approach. First, we examine app developers' discussions of Google Play's closed testing requirements on Reddit. Reddit's unique capabilities for forming and sustaining online communities make it an ideal space for software developers to share experiences, troubleshoot issues, discuss development frameworks, and explore new programming trends [12, 23, 25, 47]. However, while Reddit discussions offer valuable insights, they are often informal, fragmented, and influenced by community dynamics, making it challenging to extract structured, in-depth perspectives. To address this limitation, we conduct a survey with 14 self-identified indie Android developers, allowing us to gather targeted, firsthand accounts of their experiences with Google Play's closed testing requirements. Developer surveys are broadly used in software engineering research, particularly when individual ecosystem-specific experiences are critical for understanding nuanced challenges [4, 24, 32]. In our study, the survey will enable us to validate and expand upon themes identified in Reddit discussions, ensuring a more comprehensive and representative analysis.

3.1 Analysis of Developer Discussions on Reddit

Reddit is an online social platform that enables its users to share content and engage in discussions on a broad range of topics, from niche hobbies to global news. Discussions on Reddit are contained within subcommunites, also known as *"subreddits."* A subreddit can be described as a community that is dedicated to a particular topic. The visibility of individual discussion threads is controlled by community signals, such as the number of *"upvotes"* a thread receives and the engagement of the community, often measured by the number of comments and replies.

The abundance of technical discussions available on subreddits makes them a valuable source of information that can be utilized to understand the dynamics of software developer communities around a plethora of software engineering subjects. For instance, Li et al. [23] conducted a qualitative analysis of 207 discussion threads on Reddit to explore how privacy concerns and risky data practices arise in developers' discussions. Bouma-Sims and Acar [8] analyzed Reddit's discussions to understand how programmers incorporate user gender information into their apps. Fang et al. [12] tracked open-source communities on Reddit to model the dynamics of open-source projects' information diffusion across various social media platforms.

Similar to other types of software development, several vibrant Reddit communities have been formed around mobile app development. For instance, as of May 2024, `r/androiddev`, a subreddit that is dedicated to news for Android developers, had around 232k members, while `r/swift`, a subreddit that is dedicated to issues related to the native iOS language Swift, had close to 114k members. The discussions on these subreddits often exhibit high levels of engagement, especially on controversial topics related to mobile app development, such as cross-platform development and developer policies. Motivated by these observations, in this paper we utilize Reddit as a source of raw and timely developer feedback on Google Play's new closed testing requirements. In what follows, we describe our data collection and analysis.

Table 1. Our dataset divided by subreddits, including the number of discussion threads retrieved under each subreddit, the number of discussion threads included in our analysis, the total number of upvotes the included discussion threads received as of May 2024, the total number of comments analyzed under each subreddit, and the total number of unique Reddit users (authors) engaged in the discussions under the included threads.

Subreddit	Retrieved threads	Included threads	Total upvotes	Total comments	Total users
`r/androiddev`	10	7	550	584	372
`r/FlutterDev`	10	4	168	229	122
`r/AndroidClosedTesting`	4	1	4	11	10
`Others`	15	5	311	73	60
Total	38	17	1,033	897	564

3.1.1 Data and Qualitative Analysis. We used Google's search engine to locate posts and discussion threads related to Google Play's new closed testing requirements. Our search query started with the terms "`Google Play`" and "`Closed Testing.`" The query returned a large number of related posts in the first three Google search results pages. We identified a related post as a Reddit post that explicitly mentioned the testing requirements. However, several irrelevant posts were also retrieved (17/30 in the first three pages of Google search results), including generic posts about testing mobile apps. To narrow down the scope of the search, we added "`14 days.`" This increased the number of related posts to 27/30 in the first three pages of Google search results. We also experimented with "`20 testers,`" but the results were equally relevant. Other combinations of search terms, including the words "`Android,`" "`testing requirements,`" and "`personal accounts,`" did not retrieve any new discussion threads. Based on these results, our final Google search query was set to:

```
site:reddit.com, "Google Play" AND "closed testing" AND ("20 testers" OR "14 days")
```

The search was conducted in May 2024. Google Play imposed the new testing requirements in November 2023. Only posts after that date were included. To ensure the semantic richness of the data, only posts with three or more comments were included in the analysis. In general, posts with fewer than three comments included only ads for third-party testing services. A list of these posts is available in our supplemental data.

To conduct our analysis, three coders (authors of the paper) annotated the data. All coders hold professional degrees in software engineering with an average of 4.2 years in mobile app design and development. The first task included classifying our initial set of retrieved threads as either relevant or irrelevant. Threads that did not contain comments directly discussing the testing requirements are considered irrelevant. For example, a thread that only included exchanging testers' information was marked as irrelevant. Based on this criterion, each coder independently classified each thread, and the individual classifications were then consolidated. Of the 38 retrieved threads, 17 were considered relevant. We used Fleiss Kappa to assess the inter-rater agreement between our coders on the binary classification of the discussion threads to relevant or irrelevant [38]. Our results (Kappa = 0.75) suggested a *substantial* level of inter-rater agreement with a ***p-value*** < 0.001.

The distribution of both the retrieved and included threads over the different subreddits is shown in Table 1. The table also shows the total number of comments and upvotes the included threads under each subreddit received along with the number of unique comment authors. In general, the retrieved discussion threads are concentrated in the subreddits:

- **`r/androiddev`**: With more than 232k members, `r/androiddev` is considered the most active subreddit for Android developers. Posts on this subreddit include discussions of news, talks, tools, frameworks, and open-source projects related to native Android development. Of the ten discussion threads retrieved from this subreddit, seven were included in our analysis.

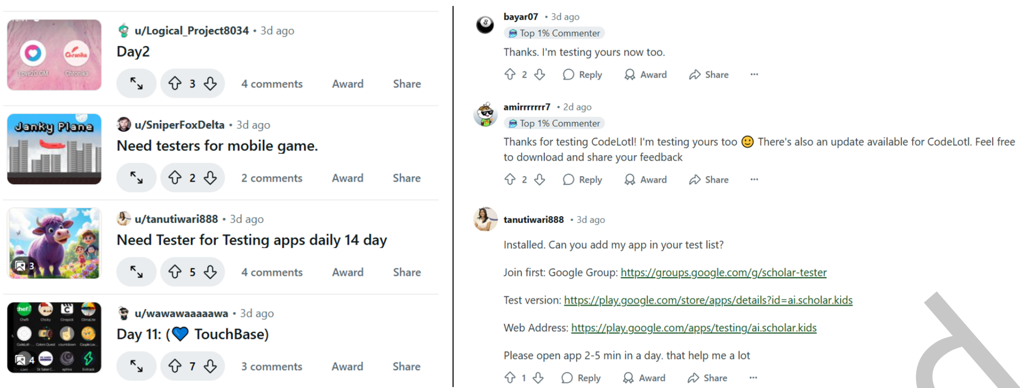

Fig. 2. Sample discussion threads posted under r/AndroidClosedTesting along with sample comments.

Table 2. The discussion threads annotated during the pilot session.

Post title	**Subreddit**	**Num. comments**
What do you guys think of the new google play updates?	`r/androiddev`	139
Google's 20 tester Android rule will be the death of Flutter	`r/FlutterDev`	96
Let's Talk About the 20 Testers Rule 10?	`r/android_devs`	10

- **`r/FlutterDev`**: This Reddit community is dedicated to news and discussions about Flutter, a cross-platform mobile app development framework that is developed and maintained by Google. As of May 2024, this subreddit had around 130k members engaging in discussions related to the Dart programming language, news about the Flutter framework, and best practices for cross-platform development. Of the ten discussion threads retrieved from this subreddit, four were included in our analysis.
- **`r/AndroidClosedTesting`**: This subreddit emerged in December 2023 to help app developers adhere to the new testing requirements. As of May 2024, the subreddit had around 1.7k members. The majority of posts were titled *"need 20 testers, will test back."* Figure 2 shows a recent sample of such posts along with sample comments that commonly appear under them.
- **`Others:`** Several related posts were scattered across other Android development subreddits but with less frequency, including `r/android_devs`, `r/IndieGaming`, and `r/LinusTechTips`. Of the 15 discussion threads retrieved from these subreddits, five were included in our analysis.

After the list of relevant threads was determined, the coders examined the comments under each thread for themes. A theme can be described as any recurring reaction, argument, plan of action, concern, or emotion the comment conveys [9]. Three discussion threads were annotated during a 3-hour pilot open coding session that included all three coders. These threads are shown in Table 2. In particular, each discussion thread was projected on a large screen. The three judges discussed the comments from the top to the bottom—a total of 245 comments, representing 27% of the total number of comments in the data. The nested nature of Reddit's comments and replies were considered to preserve context. The pilot session helped establish an initial understanding of the task, address coders' concerns, and identify the initial set of codes in the data. Codes were then consolidated into themes.

After the pilot session, three additional coding sessions were held to analyze the remaining threads, with each session covering four to five threads. Axial coding and memoing were used to track emerging codes

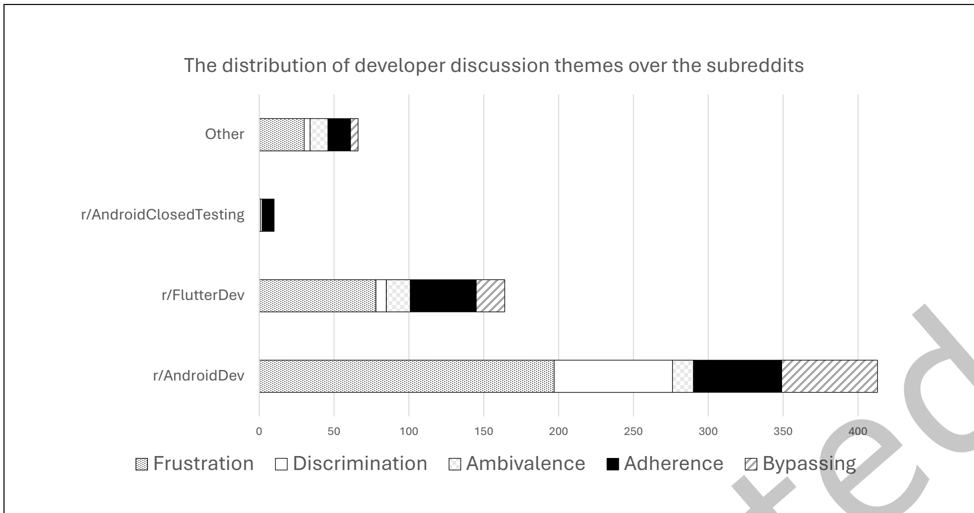

Fig. 3. The distribution of the main themes detected in indie developer Reddit discussions of Google Play's new closed testing requirements over the different subreddits.

and themes [29]. When conflicts arose during individual annotation, all three judges discussed their decisions, providing explanations for their choices. If they could not reach a consensus, they resolved disagreements through majority voting. Overall, conflicts were limited and typically involved longer comments that addressed multiple issues. For instance, the comment *"I understand why ... but 20 seems excessive"* conveys acceptance of the requirements in the first part and expresses concern in the second. Another example is the comment *"Who knows, maybe I'll have to make an LLC instead."* One judge interpreted it as sarcastic criticism of favoritism toward corporate accounts; another viewed it as a strategy to bypass the requirements; and the third saw it as an expression of frustration. After further discussion, a majority vote determined that the most appropriate label was a bypassing strategy.

A final session was held towards the end to discuss the final themes. The iterative nature of the coding process and the constant revision of codes helped us establish rigor and develop reliable themes. The annotation process concluded in six weeks. A total of 897 comments were examined across the included threads, of which 653 offered insights into developers' reactions to the testing requirements.

3.1.2 Results. Our annotation process revealed that themes in developers' discussions of the testing requirements on Reddit revolve around two main broad categories: **a)** perception of the new testing requirements, including feelings of frustration, discrimination, and ambivalence, and **b)** actionable strategies to respond to the requirements, including strategies to adhere to the requirements and strategies to bypass them. A total of 439 comments were classified under the first category and 214 comments were classified under the second. The distribution of these themes over the different subreddits is shown in Fig. 3 and a summary of the themes is provided in Table 3. The complete annotated data is provided in our supplemental material[1]. In what follows, we describe these themes.

Perception: Our qualitative analysis of developers' comments revealed that developers view Google Play's new testing requirements mainly negatively, perceiving them to be frustrating and discriminatory, with a small

[1]https://figshare.com/s/89c03a894eaaf0cd4302

Table 3. The main themes identified in developer discussions along with the number of comments classified under each theme and example comments.

Category	Theme	Description	Count	Example
Perception	Frustration	Perceiving the requirements as bureaucratic constraints that will lead to delays and longer times to market.	306	*"it's just work now with lot of unnecessary bureaucracy for the sake of it"* and *"This is an utterly senseless demand."*
	Discrimination	The belief that the requirements are designed to marginalize indie developers and dismantle their communities in favor of corporate accounts.	91	*"they're looking to kill off indie app developers."*
	Ambivalence	Mixed feelings about the requirements, ranging from acceptance to cautious optimism.	42	*"it'll create a small barrier of entry reducing malicious accounts."*
Response	Adherence	Actionable strategies to meet the requirements, such as forming tester communities, hiring third-party services, and test automation.	126	*"I made a discord server, please join and test each other apps."*
	Bypassing	Strategies to bypass the requirements, such as deploying on iOS only, migrating to web development, and forming companies.	88	*"I am starting to think about just getting into web dev, as it is a safer career path in general."*

percentage of comments praising the requirements and pointing out their benefits. These sub-themes can be described as follows:

- **Frustration:** Frustration is an emotion that occurs when an individual is blocked from completing a task or goal by reasons out of their control [53]. This emotion is the most prominent theme in the data, conveyed by comments describing the requirements as *"an absurd decision by Google"* and *"ridiculous and only aims to make indie developer's job even harder."*
  The sense of frustration in the comments mainly stems from the anticipated logistic hurdles and social pressures that indie developers may face trying to find 20 testers, which is commonly viewed as *"a steep number for solo/Indie developers."* Google recommends reaching out to family and friends to recruit testers. However, family members may not be the target user population of the app or may not be Android users. A developer on Reddit stated that *"80% in my family/friend circle use iOS."* In addition, the continuous 14-day testing requirement may not be aligned with the app's business model. For example, an app designed to connect pet owners with veterinarians is unlikely to expect or rely on daily user engagement. Imposing daily activity expectations during testing can distort the app's performance metrics and lead to misleading conclusions about its real-world usage. Such concerns appeared in comments such as *"and why would people engage every day for 20 days with an app that isn't intended to be used that way."* Our analysis also revealed that indie developers dread the social pressure they may have to endure in order to secure 20 testers. Comments such as *"I am supposed to go from coding to being a social engineer of some sort"* and *"requiring 20 testers seems like a large barrier for entry for someone with small social circles,"* highlight such an issue.
- **Discrimination:** Discrimination in the workplace can be defined as the differential or unequal treatment of employees based on their membership in certain groups [11]. Discrimination in developer communities has gained considerable attention over the past few years [7]. This line of research aims to understand the forms of bias (e.g., sexism, racism, and ageism) facing underrepresented developer groups and suggest mitigation strategies [14, 46]. Our qualitative analysis uncovered a widespread belief among indie app developers that Google Play's new testing requirements are *"anti-indie"*, only imposed to marginalize them and dismantle their communities in favor of corporate accounts. This sentiment appeared in comments such as *"they're trying to get rid of all the indie developers"* and *"they're looking to kill off indie app developers,*

hobbyists." A comment that received 25 up-votes states that, *"Google is killing the personal developers, they want to empower enterprise developers to dominate Google Play."*

Comments including discrimination concerns are frequently accompanied with concerns about the long-term impacts of the testing requirements on niche app markets. In particular, the perceived sense of favoritism towards corporate accounts may lead to a mass migration of indie developers, either to join existing corporations or to form their own. However, as existing evidence has shown, indie developers thrive in environments of freedom and autonomy [15]. Restrictions due to centralization and formalization can reduce autonomy, individual creativity, and motivation, thus depriving the ecosystem of a vital stream of entrepreneurial apps [18, 45]. This concern was echoed in comments like *"niche market apps will disappear because enterprises won't make those"* and *" looks like Google only wants enterprise apps and not indie developer apps."*

- **Ambivalence:** A smaller percentage of comments conveyed mixed emotions about the requirements, accepting and justifying them while expressing cautious optimism about their benefits. For instance, several comments point out the potential effectiveness of the requirements in countering abusive app distribution behaviors, such as releasing low-quality apps at a very high frequency. One comment argued that the new requirements *"make repeatedly submitting new apps significantly more time consuming"* and another comment stated that *"the biggest plus is it'll create a small barrier of entry reducing malicious accounts/apps."* However, such arguments are often countered by comments pointing out that malicious app distributors may develop workarounds to bypass the requirements, such as *"they [scammers] may actually create an efficient automated process for this so the only devs hurt by this are actual developers."*

Response Strategies: The second main category of themes that appeared in our dataset of Android app developers' discussions describe a set of actionable strategies to adhere to the requirements or to bypass them. Comments classified under the adherence theme contained suggestions to help indie developers find 20 testers to test their apps, including:

- **Forming tester communities:** Several comments suggest forming tester communities to help indie Android developers recruit testers. This strategy appeared in comments such as *"Indie developers can form groups and seek each other's help."* For instance, many comments include links to Discord servers and Facebook and Google groups dedicated for testers, such as *"let's help each other out. I made a discord server, please join and test each other apps."* A prominent example of such communities that appeared frequently in the comments is the subreddit `r/AndroidClosedTesting`. Indie app developers can post their Play Store links on this subreddit and members of the community can opt to test. Such posts' descriptions include direct links to the app and often a few comments, typically in the form of *"done, please test mine."* `r/20AndroidTesters` is another, but significantly smaller, subreddit that promises to make meeting Google Play's 20-tester requirement straightforward and stress-free.
- **Third-party services:** Hiring independent testing services to conduct the closed testing is suggested in multiple comments. These services mainly include third-party websites and gig workers offering 20 testers in exchange for a fee. For instance, the website Launchpad[2] provides a link for finding users and testers. App developers can subscribe and submit their apps' links in the form of a post. Other developers can then offer their services in exchange for others testing their apps. A tester can declare joining the testing group of an app by commenting on the app's page. The website claims that the service is free. However, other websites, such as 20apptester.com, provide their testing services for fees, ranging from $10 to $20 based on the size of the app.

  Another type of third-party testing services includes gig workers offering to test indie apps for a fee. Comments with references to such services often take the form of *"I will do all the leg work find and*

[2]https://flutter-launchpad.web.app/

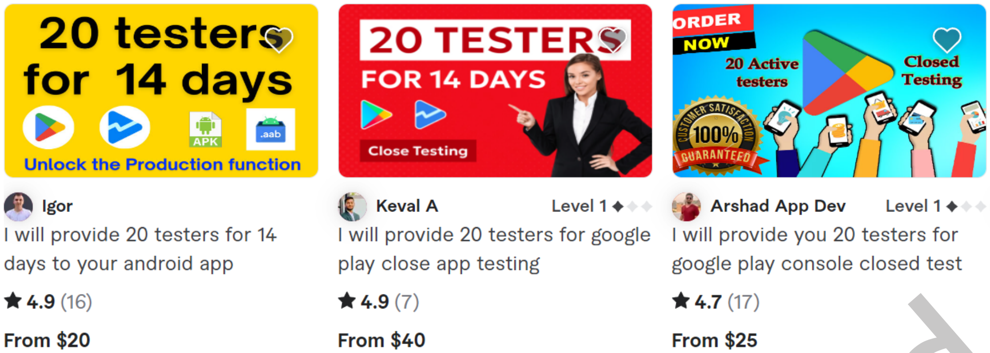

Fig. 4. Examples of a listing on Fiverr offering to provide 20 testers for a fee.

gather testers for you and do all the interfacing for a fee." In fact, a search for *20 testers* on Fiverr.com, the popular gig work platform, returns 172 listings of either individuals or groups offering their testing services. Figure 4 shows examples of such posts. The fees often range from $10 for basic plans up to $100 for premium plans promising guaranteed publishing of the app along with a report of the testing process.

- **Automation:** Test automation is also suggested as a strategy to meet the requirements. A developer states that they *"have a setup with 20 emulators, each with a unique email, to automatically test my app via a script."* However, several other comments advise against such tactics, mainly due to concerns of policy violations, such as *"I'm unsure if Google accepts this method"* and *"don't try to game the system."*

The second theme detected under the response strategies category includes discussions of strategies to bypass the closed testing requirements. Such strategies include:

- **Migration to other app distribution platforms:** This strategy appeared in numerous comments across almost all discussion threads as a potential solution to bypass Google Play's testing requirements. The Apple App Store, which currently has no such testing requirements, is frequently suggested as a migration destination, appearing in comments such as *"time to learn Swift"* and *"I turned to iOS and I hate the Google ecosystem lately."* However, suggestions of migration to the App Store are commonly associated with concerns about the higher annual fees for Apple developer accounts ($99 per year in comparison to a one time $25 payment for Google Play) as well as the relatively higher costs of the infrastructure needed to develop iOS apps. Such concerns appeared in comments such as *"my main issue currently is not having a Mac"* and *"nope no testing required. Does require a DUNS number, mac, and $100/year though."*
  Third-party Android platforms, such as F-Droid, and open-source platforms, such as GitHub, were also suggested as possible migration destinations, appearing in comments such as *"It's simple, we migrate to F-Droid"* and *"just put the APKs on GitHub."* Such a tactic is commonly known as *sideloading*, where an app's binaries are distributed to users' devices from outside official app stores, bypassing their restrictions. However, such comments were accompanied by concerns about the outreach of third-party platforms in comparison to Google Play, such as *"I wonder if 3rd party app stores will fill the void for smaller apps."* Security concerns were also raised about the long-term impacts of bypassing the security requirements of Apple and Google on the app's ability to survive.
- **Switching to Web development:** Our analysis revealed that, due to the complications associated with mobile app development and the increasing platform gatekeeping, web development is increasingly perceived as an alternative with *"more control and less bureaucracy."* This sentiment appeared in comments

such as *"I am starting to think about just getting into web dev, as it is a safer career path in general"* and *"take the web dev route, the freedom of the internet will never be taken away from us."*

- **Forming companies:** Google Play's closed testing requirements do not apply to corporate accounts. Several comments indicated that indie developers operating under personal accounts were considering forming their own companies, especially Limited Liability Companies (LLCs), to bypass the requirements. This strategy appeared in comments such as *"because of this I am registering a UK LTD company through an agent"* and *"maybe I'll have to make an LLC instead."* However, similar to other strategies, concerns were raised about the legal complications and high fees associated with forming organizations, especially outside of the U.S. market. This concern appeared in comments such as *"I'm not willing to set up an LLC and deal with all the tax complications and other legal hassles just to get around this rule."*

3.2 Developer Survey

We conducted a survey with 14 self-proclaimed indie Android app developers to gain further insights into their perception of Google Play's closed testing requirements. Developer surveys are commonly used in software engineering research to generate elaborate discussions of subjects where individual ecosystem-specific experiences are critical [24, 32, 40]. In what follows, we describe our survey setup, participant response analysis, and main findings.

3.2.1 Survey Structure and Analysis. To recruit our survey participants, we reached out to several app developer communities on Reddit. Our selection criteria included Android app developers who operated under personal developer accounts, spoke English, and either recently passed Google Play's closed testing requirements, or were seeking to adhere to them by the time of the survey. To recruit participants from Reddit, private messages were sent out to posters on the subreddits identified earlier. A principal investigator committed to act as a tester for 14 days in exchange for participating in the survey. To identify potential survey participants, we examined the comments under each sub-thread, looking for indications that a developer was currently undergoing or had recently completed the approval process. We searched for phrases like *"I'm on day one of 14 days of closed testing"* and *"I have 17 testers."* This process yielded 46 accounts that met our inclusion criteria, of which 30 had direct messaging (chat) enabled on Reddit.

A total of 30 messages were sent and 10 responses were received. However, only eight participants from Reddit were considered; a participant never sent their response while another participant demanded daily videos of their app being used in exchange for completing the survey, which was not feasible. To enhance the diversity of our study participant sample, we leveraged our connections with a local technology incubator that provides support for technology startups and solopreneurs. A total of 24 developers who worked at the incubator were contacted through email. Six developers met our inclusion criteria, raising the number of our survey participants to 14. Our survey consisted of three open-ended questions, followed by four demographic questions:

Survey questions:
- Why did you choose Google Play to publish your app?
- What are your plans to comply with the new testing requirements?
- What is your assessment of such requirements?

Demographic questions:
- What is the main application domain of your app?
- What country do you operate from?
- How old are you (optional)?
- How many years of experience do you have as a mobile app developer?

Each participant had to sign a consent form before completing the survey. They were also informed that they could opt out of completing the survey at any time and that no personally identifiable information would be disclosed. All participants sent their responses within two weeks.

The same three coders from the first study conducted an inductive and iterative thematic analysis of our participants' responses. Two responses from a U.S. resident and a non-U.S. resident survey participants were initially coded in a group pilot session. Generic themes quickly emerged in the data as most responses were brief. Each researcher then coded the rest of the surveys individually [29]. Coding stopped when saturation was reached; no more response categories emerged. A group session was then held to consolidate individual codings and reconcile discrepancies [9]. Given the direct nature of our questions, conflicts were limited and only pertained to labeling issues (naming the theme category). Once the labels were determined, coding proceeded with no conflicts. The analysis concluded in two days. Given the categorical nature of the analysis, a Fleiss' Kappa analysis was conducted to assess the level of agreement among raters for each question. The analysis yielded Kappa values of 0.67, 0.67, and 0.61 for questions 1, 2, and 3 respectively, indicating a substantial agreement among the coders [38]. The results were statistically significant for all three questions with a ***p-value*** < 0.001. The fully annotated survey responses and individual codings are included in the supplemental material. A summary of our survey participants' responses is provided in Table 4. Our 14 survey participants are given the codes P_1, P_2, P_3, ..., P_{14}.

3.2.2 Results. Our analysis of the responses to our demographic questions shows that our survey participants develop apps in various domains, including social media, sharing (gig) economy, gaming, and productivity. Non-U.S. resident participants are from India, Pakistan, Portugal, and Jordan. Our participants are on the younger side (average age is 24.2) and they have an average of 2.4 years of experience in mobile app development. In what follows, we describe the themes identified under each of our research questions:

- **Q_1: Why did you choose Google Play to publish your app?**
  Five (N = 5) of our survey participants listed the relatively high fees of the Apple App Store as the reason for choosing Google Play to host their apps. A few survey participants (N = 3) listed the perceived freedom and flexibility that Google provides in comparison to Apple as motivations for targeting Google Play. P_4 stated that *"my monetization strategy and ad revenue model is not approved by Apple."* P_5 indicated that *"Google is more flexible. Everyone knows that."* Seven (N = 7) non-U.S. resident survey participants indicated that Android phones were significantly more popular in their countries of operation, thus it made sense to target Google Play. Three (N = 3) survey participants indicated that they were using cross-platform frameworks to develop their apps. This prompted them to deploy on both app stores. P_{10} stated that *"I picked up Flutter and its cross platform, I thought I would deploy on Android too."*

Table 4. Survey responses and the themes identified under the open-ended questions $\mathbf{Q_1}$, $\mathbf{Q_2}$, and $\mathbf{Q_3}$.

Participant	Country	Domain	Q_1	Q_2	Q_3
$\mathbf{P_1}$	U.S.	Social Media	Cross-platform development	Reddit and friends and family	Frustrating and discriminatory
$\mathbf{P_2}$	U.S.	Sharing Economy	Fees	Target users	Useful but frustrating
$\mathbf{P_3}$	U.S.	Sharing Economy	Cross-platform development	Friends and family, target users, and third-party	Frustrating and discriminatory
$\mathbf{P_4}$	U.S.	Productivity	Fees and flexibility	Reddit and third-party	Frustrating and discriminatory
$\mathbf{P_5}$	India	Productivity	Fees and flexibility	Reddit	Frustrating and discriminatory
$\mathbf{P_6}$	India	Sharing Economy	Market demand	Reddit	Frustrating
$\mathbf{P_7}$	Pakistan	Education	Market demand	Reddit and third-party	Frustrating and discriminatory
$\mathbf{P_8}$	Pakistan	Sharing Economy	Fees and market demand	Friends and family, Reddit, and third-party	Frustrating and discriminatory
$\mathbf{P_9}$	Pakistan	Lifestyle	Fees, flexibility, and market demand	Reddit and third-party	Frustrating and discriminatory
$\mathbf{P_{10}}$	Portugal	Travel	Cross-platform development and market demand	Reddit	Useful but frustrating and discriminatory
$\mathbf{P_{11}}$	Portugal	Gaming	Market demand	Reddit and friends and family	Frustrating and discriminatory
$\mathbf{P_{12}}$	Jordan	Gaming	Market demand	Reddit and friends and family	Frustrating
$\mathbf{P_{13}}$	U.S.	Lifestyle	Cross-platform development and market demand	Reddit and friends and family	Frustrating
$\mathbf{P_{14}}$	U.S.	Sport	Market demand	Reddit and friends and family	Frustrating and discriminatory

- **Q_2: What are your plans to comply with the new testing requirements?**
The majority of our survey participants (N = 12) indicated utilizing Reddit to recruit testers. However, several participants (N = 5) pointed out that their attempts at securing testers on Reddit failed multiple times. P_9 stated that *"testers would randomly stop testing. We had to start over two times."* P_6, who also participated in testing other apps, stated that in some cases, the requirements were tedious, such as requiring a daily proof of app usage in the form of a video recording. Four (N = 4) survey participants indicated that they opted for paid services on Fiverr, mainly as a backup plan. P_8 indicated that they feared getting banned due to Google's strict policies on manipulation, stating that *"to be honest I was worried this was not going to end well for me"* and *"I wasn't even sure such services were legal or if I was going to get banned."* Five (N = 5) survey participants indicated that they asked their friends and family members to test their apps, and only two (N = 2) indicated reaching out to their target user population. Three (N = 3) of our U.S. resident survey participants cited the complexities and fees associated with creating a company as a reason for not considering that option. P_1 stated that *"I tried to create an LLC*

and I was confused and lost" and P_2 stated that *"I'm not even sure my app would work, why would I get tangled with the IRS at this point."* Two survey participants (N = 2) who are residents of Pakistan pointed out that indie developers in their country were looking into forming some sort of virtual companies to provide protection against any future constraints from app stores on personal developers. P_4 indicated that they were seriously considering shutting down their app, stating that *"I just cannot find 20 people and ask them for such a long commitment,"* while P_{10} indicated that they considered not deploying on Google Play, stating that *"at some point I thought I would go only on Apple."*

- **Q_3: What is your assessment of such requirements?**
  Our developer survey showed that only two survey participants (N = 2) perceived the new testing requirements positively. P_{10} stated that *"It could stop spam apps especially video games."* However, all of our survey participants (N = 14) confirmed that the new testing requirements were frustrating, using words such as *tedious*, *annoying*, and *unnecessary*. Ten participants (N = 10) perceived the requirements as discriminatory and an attempt from Google to eliminate the indie app market. This sentiment appeared in responses such as *"they want to make more space for corporations"* and *"Google doesn't look favorably on indie developers. They want us out."* P_{11} raised concerns about the impact of the testing requirements on indie developers outside of the U.S., stating that *"it is so hard to establish an LLC outside of the U.S. These requirements harm developers outside of the U.S. the most."*

4 DISCUSSION AND IMPLICATIONS

The study of platform policies, such as privacy policies [5], anti-discrimination policies [44], and general developer policies [24], has gained increasing attention in recent years. This line of research aims to evaluate the structure and format of such policies, identify their limitations and unintended side effects, and outline best practices for compliance throughout the different phases of the software engineering process. Our work in this paper extends this line of research by focusing on Google Plays' closed testing requirements. In particular, our study answers two main research questions:

RQ_1: How do indie app developers perceive the new testing requirements? Our qualitative analysis of Reddit discussions and survey responses reveals that indie app developers choose Google Play for several reasons, including its perceived flexibility, lower fees, and the widespread use of Android phones—particularly in markets outside the U.S. However, our findings show that the platform's closed testing requirements are often viewed as frustrating and discriminatory. Developers see them as unnecessary barriers to market entry that could threaten the survival of niche apps. Many also express concern about the inflexibility of these requirements and their lack of accommodation for diverse app business models.

RQ_2: How do indie app developers navigate the testing requirements? Our analysis reveals that developers rarely engage their target user base for app testing. Instead, they rely on family and friends, Reddit testing communities, or third-party services. However, concerns are raised about the legitimacy of these services and the risks of associating with previously banned accounts operating under these services. Forming a company is seen as a less feasible workaround, largely because of the complexities involved in setting up and managing a business, particularly outside the U.S. market. In what follows, we discuss the implications of these findings on the state of practice and research.

4.1 Implications for Google Play

Google states that its testing requirements for indie apps are intended to *"help all developers deliver high-quality apps"* and describes them as a strategy to make testing *"an integral part of the app development process."* However, our analysis shows that, as they currently stand, the requirements are limited by design, introducing barriers to adaptation and compliance. To address these limitations, we suggest the following:

- **Adaptive testing requirements:** The testing requirements are enforced as a *one-size-fits-all*. However, one of the issues that frequently appeared in our developer survey and Reddit comment analysis is the misalignment between the 14-day constraint and apps' business models. As mentioned earlier, not all apps expect daily engagement from their end-users. A suggested strategy to address these limitations is to alter the testing requirements to adapt to different business models. App developers can declare the specific application domain of their app (e.g., ride-hailing, investing, mental health) as well as the expected engagement frequency. The 20 testers can then engage with the app 14 times but within an estimated period of time that is proportional to the expected engagement rate. Relaxing the 14-day constraint can substantially reduce the pressure on both developers and testers and can provide more realistic app usage scenarios.
- **Google verified testers:** Our findings revealed a previously undocumented phenomenon in response to platform policies, characterized by the emergence of third-party testing services and specialized developer communities dedicated to closed testing. However, major concerns are raised about the unverified tactics of such intermediary services. For instance, Google Play frequently cites association with terminated developer accounts as a reason for account termination. These concerns appeared in comments such as *"what if they used terminated accounts? is that going to get me terminated by association?"* Other concerns included the potential exploitation of underpaid labor by what is described in a Reddit comment as *"sweatshop software houses."*
  To mitigate such concerns, Google can crowd-source the testing process through some sort of a reward program that testers and early adopters can subscribe to. Testers can then be rewarded based on their involvement in testing apps and the quality of feedback they provide. Indie app developers can recruit testers from the pool of verified testers. Google provides similar reward programs for its Google Reviews platform which relies on the crowd to rate and review businesses. Active Google reviewers are often rewarded with various forms of incentives, such as early access to businesses or new Google products [21]. By taking the initiative to regulate the testing market, Google can effectively curb the stream of unverified third-party services, thus preventing predatory intermediary organizations from exploiting indie app developers through in-perpetuity contracting or unfair revenue splits [27, 41].

4.2 Implications for Indie Developers

Our findings show that indie app developers often perceive Google Play's closed testing requirements as logistical barriers to market entry. These concerns stem from limited resources and the inherently agile nature of indie production cycles, where rigid testing mandates can be difficult to accommodate. To help indie developers effectively integrate the requirements into their workflows, we propose the following strategies:

- **Test-driven ideation:** Strategizing and planning for the closed testing stage should shift from being an after-the-fact concern to a pre-ideation task. Indie developers should seek validation to their ideas before implementation. This involves reaching out to their target user population to get a sense of their needs. By defining a clear business hypothesis, developers can validate the viability of their ideas early in the process and make necessary pivots before implementation. If the idea is effectively validated and motivated, finding 20 users for closed testing should be manageable. In fact, this recommendation appeared in several Reddit comments such as *"if you can't find anybody interested in your app that's a feedback for you as well."*
- **Community building:** Once the idea is validated, indie developers should start an effort to build a community around the idea. This effort should continue throughout the implementation process. Social media channels and specialized communities on platforms such as Reddit can be utilized to identify potential community members. For example, a survey participant indicated that they were developing an app to help people with stuttering problems. For such an app, the `r/stutter` subreddit, which has more

than 19k members, can be a good start. The target community members should be constantly engaged by informing them about the status of the project. Once the app is ready for testing, members of the community can be recruited to conduct the test.

- **Lean engineering:** As mentioned earlier, indie development is often driven by agility and efficiency. The new testing requirements are perceived as barriers that can slow down the time to market. This limitation can be addressed by integrating more lean software engineering practices into the development process. In particular, any aspect of the process that does not lead to an immediate customer value is considered a waste and is eliminated [34, 39]. This could include, for instance, releasing a prototype or minimum viable product (MVP) for closed testing purposes rather than a fully developed product, thus accelerating the time to market and reducing the cost of failure [39].

4.3 Implications for Research and Practice

Our findings in this paper provide a foundation for future studies to explore the short and long-term impacts of Google Play's testing requirements on policy, inclusion, and innovation in the mobile app market. Based on our analysis, we identify the following directions for future research:

- **Impact on app quality:** Future research should independently investigate the effectiveness of the new testing requirements in curbing the influx of low-quality apps on Google Play. This analysis can be conducted through longitudinal studies that observe the quality of indie apps over extended periods of time [26]. In addition, the effectiveness of the new requirements should be systematically compared to other forms of quality control, such as raising developer fees to act as a financial entry barrier to malicious apps. Independent research has the potential to increase transparency and reduce mistrust between indie app developers and Google Play. Furthermore, systematic evidence can help competitor app stores make informed decisions regarding their developer policies and predict their impacts on talent retention and movement between app stores.
- **Impact on indie developers:** Indie app developers are driven by a profound quest to change the landscape of computing with their creative and innovative apps [22]. Our results showed that, with little to no immediate return on investment, any form of discouragement or platform gatekeeping could lead to abandoning such a quest [41]. Given these observations, future research should examine the impact of the testing requirements on indie app developers' ability and willingness to contribute to the mobile app market. This line of research should also investigate how indie developers' response strategies, such as migrating to other platforms and forming organizations, can affect their creativity and innovation, particularly in niche market segments. It is crucial to also consider the diverse demographics of indie developers (e.g., gender, nationality, application domain, etc.) to identify the best strategies for mitigating the impact of testing requirements on various marginalized developer groups.
- **Third-party services:** Our study identified a trend of third-party organizations that has emerged to act as downstream intermediaries between app developers and Google Play [27]. Organizational research has shown that the formation of such services in other domains, like YouTube Multi-channel Networks (MCNs), can significantly influence creativity and fairness. These effects can range from the exploitation of small creators to having full control over innovation and creativity in the app market [27, 41]. Future research should investigate the tactics of such services and their impacts before they become reality in the mobile app market.

5 LIMITATIONS

The study presented in this paper has several limitations that could potentially jeopardize the validity of the results. A limitation may arise from the fact that our analysis of online developer feedback is limited to 17 discussion threads from Reddit. Other platforms such as Stack Overflow, Twitter, and Hacker News are also important sources of developer discussions and could offer additional insights [2, 54]. Furthermore, since participation on Reddit is pseudonymous, it is often not possible to obtain the exact demographic characteristics of participants [35]. However, as noted in our data section, Reddit is broadly used in software engineering research [16, 20, 35, 48]. The app developer communities on Reddit are relatively large and active, providing timely, in-depth, and unfiltered feedback on current challenges faced by developers. In addition, the structure of Reddit encourages users to engage in existing discussions rather than create new threads, which explains why our selected discussion threads were particularly rich in meaningful discourse. Nonetheless, we acknowledge the fact that, due to the unique characteristics of Reddit communities, discussion threads on Reddit can act like echo chambers, thus a confirmation bias may have influenced our findings [35].

Another limitation of our study may arise from the fact that we surveyed 14 developers. This sample size can limit the generalizability of our findings to other demographics of app developers. However, due to the nature of the problem, our inclusion criteria significantly restricted the size of our population. We mitigated this threat by recruiting a diverse sample of indie developers in terms of geographic location and application domain. We further recruited six subjects from outside the Reddit subject pool, thus enhancing the generalizability of our findings to the broader population of indie developers. To minimize bias, our subjects included developers who had already passed the requirements and developers who were actively seeking to adhere to them by the time of survey. In general, both of our studies can be sufficient to derive meaningful insights—especially since the survey results aligned with our Reddit discussion analysis, reinforcing the validity of our findings.

A potential threat to internal validity arises from the manual annotation of Reddit comments. While manual labeling is a common practice in qualitative research [29], the choice of coding labels may vary depending on the annotators. For example, we use *discrimination* to categorize comments that reflect favoritism, targeting, or differential treatment. This label is based on a recent definition of discrimination from the Oxford Handbook of Workplace Discrimination [11]. Annotators were instructed to refer to such definitions to annotate the comments. To mitigate the influence of extraneous factors such as boredom and mental fatigue, we imposed no time constraints on annotators. These measures helped maintain data integrity, leading to a low conflict rate between annotators. Overall, independent replications of our study would yield similar findings.

6 CONCLUSION

In this paper, we investigated the impact of Google Play's new closed testing requirements for apps submitted by personal accounts on indie app developers. Using qualitative research methods, we analyzed Android developers' discussions on Reddit and conducted a survey with 14 indie app developers to gauge their perception of the new requirements. Our results indicated that indie app developers perceived the new requirements as inflexible, discriminatory, and frustrating, imposing bureaucratic and logistical barriers that could potentially hinder their ability to innovate and compete in the app market. These concerns are summed up in the comment:

> *"As a developer, the new policy requiring at least 20 testers for two weeks feels overly burdensome. This isn't just about the challenge of finding reliable testers, but the added time and resource constraints it places on already stretched development cycles. For small teams or solo devs, this could mean significant delays or even the shelving of innovative projects. We need a policy that ensures quality without stifling creativity and agility in the app development process."*

Our analysis has also uncovered strategies that developers may employ to meet the closed testing requirements, such as leveraging their own online communities for testing or hiring third-party services and gig workers. Based on our findings, we proposed strategies to mitigate the impact of the testing requirements on indie app developers, such as providing flexible requirements that can adapt to various application domains and taking the initiative to regulate the testing market. In addition, we proposed several guidelines to help indie developers comply with Google Play's testing requirements while maintaining their agility and efficiency, such as adapting more lean software development practices, community building, and test-driven ideation. Finally, we suggested several directions for future research to better understand the effects of quality assurance policies on app developers and users, as well as propose strategies to mitigate their impacts on innovation and fairness in the mobile app market.

REFERENCES

[1] Afnan A. Al-Subaihin, Federica Sarro, Sue Black, Licia Capra, and Mark Harman. 2021. App Store Effects on Software Engineering Practices. *IEEE Transactions on Software Engineering* 47, 2 (2021), 300–319.

[2] Alessia Antelmi, Gennaro Cordasco, Daniele De Vinco, and Carmine Spagnuolo. 2023. The Age of Snippet Programming: Toward Understanding Developer Communities in Stack Overflow and Reddit. In *Companion Proceedings of the ACM Web Conference*. 1218–1224.

[3] Apple. 2023. 2023 App Store Transparency Report. https://www.apple.com/legal/more-resources/docs/2023-App-Store-Transparency-Report.pdf. Accessed: 2024-05-05.

[4] Terese Besker, Antonio Martini, Rumesh Edirisooriya Lokuge, Kelly Blincoe, and Jan Bosch. 2018. Embracing Technical Debt, from a Startup Company Perspective. In *The International Conference on Software Maintenance and Evolution*. 415–425.

[5] Jaspreet Bhatia and Travis Breaux. 2018. Semantic incompleteness in privacy policy goals. In *The IEEE International Requirements Engineering Conference*. 159–169.

[6] Markus Borg, Vahid Garousi, Anas Mahmoud, Thomas Olsson, and Oskar Stålberg. 2020. Video Game Development in a Rush: A Survey of the Global Game Jam Participants. *IEEE Transactions on Games* 12, 3 (2020), 246–259.

[7] Amiangshu Bosu and Kazi Zakia Sultana. 2019. Diversity and Inclusion in Open Source Software (OSS) Projects: Where Do We Stand?. In *The International Symposium on Empirical Software Engineering and Measurement*. 1–11.

[8] Elijah Bouma-Sims and Yasemin Acar. 2023. Beyond the Boolean: How Programmers Ask About, Use, and Discuss Gender. *The ACM on Human-Computer Interaction* 28 (2023), 1–31.

[9] Virginia Braun and Victoria Clarke. 2006. Using Thematic Analysis in Psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.

[10] Michael Cao, Khaled Ahmed, and Julia Rubin. 2022. Rotten Apples Spoil the Bunch: An Anatomy of Google Play Malware. In *The International Conference on Software Engineering*. 1919–1931.

[11] Adrienne Colella and Eden King. 2017. *The Oxford Handbook of Workplace Discrimination*. Oxford University Press.

[12] Hongbo Fang, Bogdan Vasilescu, and James Herbsleb. 2023. Understanding information diffusion about open-source projects on Twitter, HackerNews, and Reddit. In *The IEEE International Conference on Cooperative and Human Aspects of Software Engineering*. 56–67.

[13] Guo Freeman, Jeffrey Bardzell, Shaowen Bardzell, and Nathan McNeese. 2020. Mitigating Exploitation: Indie Game Developers' Reconfigurations of Labor in Technology. *Proceedings of the ACM on Human-Computer Interaction* 4 (2020), 1–23.

[14] Guo Freeman, Lingyuan Li, Nathan Mcneese, and Kelsea Schulenberg. 2023. Understanding and Mitigating Challenges for Non-Profit Driven Indie Game Development to Innovate Game Production. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.

[15] Guo Freeman and Nathan Mcneese. 2019. Exploring Indie Game Development: Team Practices and Social Experiences in A Creativity-Centric Technology Community. In *Computer Supported Cooperative Work*. 723–748.

[16] Robert P Gauthier, Mary Jean Costello, and James R Wallace. 2022. "I Will Not Drink With You Today": A Topic-Guided Thematic Analysis of Addiction Recovery on Reddit. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–17.

[17] Google. 2023. App Testing Requirements for New Personal Developer Accounts. https://support.google.com/googleplay/android-developer/answer/14151465?hl=en. Accessed: 2024-05-05.

[18] Giles Hirst, Daan Van Knippenberg, Chin-hui Chen, and Claudia A. Sacramento. 2011. How Does Bureaucracy Impact Individual Creativity? A Cross-Level Investigation of Team Contextual Influences on Goal Orientation–Creativity Relationships. *Academy of Management Journal* 54, 3 (2011), 624–641.

[19] Thomas Huber, Thomas Kude, Jan Lepoutre, and Julien Malaurent. 2023. *Chapter 12: Platform Governance as a Social Movement*. Edward Elgar Publishing.

[20] Tobias Kauer, Marian Dörk, Arran L. Ridley, and Benjamin Bach. 2021. The Public Life of Data: Investigating Reactions to Visualizations on Reddit. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.

[21] Hanlin Li and Brent Hecht. 2021. 3 Stars on Yelp, 4 Stars on Google Maps: A Cross-Platform Examination of Restaurant Ratings. *Proceedings of the ACM on Human-Computer Interaction* 4, 254 (2021), 1–25.

[22] Lingyuan Li, Guo Freeman, and Nathan J. McNeese. 2022. Channeling End-User Creativity: Leveraging Live Streaming for Distributed Collaboration in Indie Game Development. *Proceedings of the ACM on Human-Computer Interaction* 6, 282 (2022), 1–28.

[23] Tianshi Li, Elizabeth Louie, Laura Dabbish, and Jason I. Hong. 2021. How Developers Talk About Personal Data and What It Means for User Privacy: A Case Study of a Developer Forum on Reddit. *The ACM on Human-Computer Interaction* 220 (2021), 1–28.

[24] Tianshi Li, Kayla Reiman, Yuvraj Agarwal, Lorrie Cranor, and Jason Hong. 2022. Understanding Challenges for Developers to Create Accurate Privacy Nutrition Labels. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–24.

[25] Bin Lin, Nathan Cassee, Alexander Serebrenik, Gabriele Bavota, Nicole Novielli, and Michele Lanza. 2022. Opinion Mining for Software Development: A Systematic Literature Review. *The ACM Transactions on Software Engineering and Methodolgy* 31, 3 (2022), 1–41.

[26] Fuqi Lin, Haoyu Wang, Liu Wang, and Xuanzhe Liu. 2021. A Longitudinal Study of Removed Apps in iOS App Store. In *The International World Wide Web Conference (WWW)*. 1435–1446.

[27] Ramon Lobato. 2016. The Cultural Logic of Digital Intermediaries: YouTube Multichannel Networks. *Convergence* 22, 4 (2016), 348–360.

[28] Iren Mazloomzadeh, Gias Uddin, Foutse Khomh, and Ashkan Sami. 2024. Reputation Gaming in Crowd Technical Knowledge Sharing. *The ACM Transactions Software Engineering and Methodolgy* (2024).

[29] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and Inter-rater Reliability in Qualitative Research: Norms and Guidelines for CSCW and HCI Practice. *Proceedings of the ACM on Human-Computer Interaction* 3 (2019), 1–23.

[30] Naeem Moosajee and Pejman Mirza-Babaei. 2016. Games User Research (GUR) for Indie Studios. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 3159–3165.

[31] Ruchi Panchanadikar, Guo Freeman, Lingyuan Li, Kelsea Schulenberg, and Yang Hu. 2024. "A New Golden Era" or "Slap Comps": How Non-Profit Driven Indie Game Developers Perceive the Emerging Role of Generative AI in Game Development. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–7.

[32] Maulishree Pandey, Vaishnav Kameswaran, Hrishikesh V. Rao, Sile O'Modhrain, and Steve Oney. 2021. Understanding Accessibility and Collaboration in Programming for People with Visual Impairments. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 30.

[33] Craig Peters. 2023. Apple and Google App Stores v. Developers. *Washington University Global Studies Law Review* 22 (2023), 87.

[34] Mary Poppendieck and Michael A. Cusumano. 2012. Lean Software Development: A Tutorial. *IEEE Software* 29, 5 (2012), 26–32.

[35] Nicholas Proferes, Naiyan Jones, Sarah Gilbert, Casey Fiesler, and Michael Zimmer. 2021. Studying Reddit: A Systematic Overview of Disciplines, Approaches, Methods, and Ethics. *Social Media + Society* 7, 2 (2021).

[36] Yixin Qiu, Anandasivam Gopal, and Il-Horn Hann. 2017. Logic Pluralism in Mobile Platform Ecosystems: A Study of Indie App Developers on the iOS App Store. *Information Systems Research* 28, 2 (2017), 225–249.

[37] Mahmudur Rahman, Mizanur Rahman, Bogdan Carbunar, and Duen Horng Chau. 2017. Search Rank Fraud and Malware Detection in Google Play. *IEEE Transactions on Knowledge and Data Engineering* 29, 6 (2017), 1329–1342.

[38] J Richard Landis and Gary G Koch. 1977. An Application of Hierarchical Kappa-type Statistics in the Assessment of Majority Agreement among Multiple Observers. *Biometrics* 33, 2 (1977), 363–374.

[39] R. Pito Salas. 2017. Teaching Entrepreneurship in Computer Science: Lessons learned. In *IEEE Frontiers in Education Conference*. 1–7.

[40] Sk Golam Saroar and Maleknaz Nayebi. 2023. Developers' Perception of GitHub Actions: A Survey Analysis. In *International Conference on Evaluation and Assessment in Software Engineering*. 121–130.

[41] Michael L. Siciliano. 2023. Intermediaries in the Age of Platformized Gatekeeping: The Case of YouTube "Creators" and MCNs in the U.S. *Poetics* 97 (2023), 101748.

[42] Caio Steglich, Sabrina Marczak, Rodrigo Pereira dos Santos, Luiz Guerra, Luiz Mosmann, Marina Moreira, and Marcelo Perin. 2015. Factors that Affect Developers Decision to Participate in a Mobile Software Ecosystem. *Journal of Systems and Software* 205 (2015), 111808.

[43] Alexander Styhre. 2020. *Who Is an Indie Developer? Sorting Out the Categories*. Springer International Publishing, 107–128.

[44] Miroslav Tushev, Fahimeh Ebrahimi, and Anas Mahmoud. 2021. Analysis of Non-Discrimination Policies in the Sharing Economy. In *The International Conference on Software Maintenance and Evolution*. 104–113.

[45] Frank Ulrich and Shegaw Anagaw Mengiste. 2014. The Challenges of Creativity in Software Organizations. In *Creating Value for All Through IT*, Birgitta Bergvall-Kåreborn and Peter Axel Nielsen (Eds.). 16–34.

[46] Bogdan Vasilescu, Daryl Posnett, Baishakhi Ray, Mark G.J. van den Brand, Alexander Serebrenik, Premkumar Devanbu, and Vladimir Filkov. 2015. Gender and Tenure Diversity in GitHub Teams. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 3789–3798.

[47] Daniel Votipka, Mary Nicole Punzalan, Seth M. Rabin, Yla Tausczik, and Michelle L. Mazurek. 2021. An Investigation of Online Reverse Engineering Community Discussions in the Context of Ghidra. In *IEEE European Symposium on Security and Privacy (EuroS&P)*. 1–20.

[48] Franz Waltenberger, Simon Höferlin, and Michael Froehlich. 2023. Reddit Insights: Improving Online Discussion Culture by Contextualizing User Profiles. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–6.

[49] Haoyu Wang, Hao Li, and Yao Guo. 2019. Understanding the Evolution of Mobile App Ecosystems: A Longitudinal Measurement Study of Google Play. In *The International World Wide Web Conference (WWW)*. 1988–1999.

[50] Haoyu Wang, Hao Li, Li Li, Yao Guo, and Guoai Xu. 2018. Why are Android Apps Removed from Google Play? A Large-scale Empirical Study. In *The International Conference on Mining Software Repositories*. 231–242.

[51] Quan Wang, Beibei Li, and Param Vir Singh. 2018. Copycats vs. Original Mobile Apps: A Machine Learning Copycat-Detection Method and Empirical Analysis. *Information Systems Research* 29, 2 (2018), 273–291.

[52] Wei Wang, Yuanyuan Li, Xing Wang, Jiqiang Liu, and Xiangliang Zhang. 2018. Detecting Android Malicious Apps and Categorizing Benign Apps with Ensemble of Classifiers. *Future Generation Computer Systems* 78 (2018), 987–994.

[53] Gloria Washington. 2013. *Does User Frustration Really Decrease Task Performance?* Taylor & Francis.

[54] Grant Williams and Anas Mahmoud. 2018. Modeling User Concerns in the App Store: A Case Study on the Rise and Fall of Yik Yak. In *International Requirements Engineering Conference*. 64–75.

[55] Anu Yamunan, Khawaja Shams, and Mohet Saxena. 2023. How We Fought Bad Apps and Bad Actors in 2022. https://security.googleblog.com/2023/04/how-we-fought-bad-apps-and-bad-actors.html. Accessed: 2024-05-05.

[56] Wenhan Zhu, Sebastian Proksch, Daniel M. German, Michael W. Godfrey, Li Li, and Shane McIntosh. 2024. What is an App Store? The Software Engineering Perspective. *Empirical Software Engineering* 29, 35 (2024), 1–39.